

```

1  #include <stdio.h>
2  #include <regex.h>
3
4  int main() {
5      regex_t regex;
6      char S[100];      // Input string
7      int result;
8
9      // Predefined regular expression
10     char R[] =
11     "^ (?.*[0-9]) (?.*[^a-zA-Z0-9]).{6,}$";
12     /*
13     Regex Condition Explanation:
14
15     ^                                -> Start of the
16     string
17     (?.*[0-9])                      -> Must contain
18     at least one digit
19     (?.*[^a-zA-Z0-9]) -> Must contain
20     at least one special character
21     .{6,}                            -> Minimum
22     length of the string should be 6 characters
23     $                                -> End of the
24     string
25
26     Overall Condition:
27     The input string must be at least
28     6 characters long,
29     contain at least one digit,
30     and contain at least one special
31     symbol.
32
33     */
34
35     // Read input string using scanf
36     printf("Enter input string: ");
37     scanf("%s", S);
38
39     // Compile the regular expression
40     result = regcomp(&regex, R, REG_EXTENDED);
41     if (result != 0) {
42         printf("Regex compilation failed\n");
43         return 1;
44     }
45 }

```

```
35     }
36
37     // Match input string with predefined regex
38     result = regexec(&regex, S, 0, NULL, 0);
39
40     if (result == 0) {
41         printf("Match Found\n");
42     } else {
43         printf("No Match\n");
44     }
45
46     // Free allocated memory
47     regfree(&regex);
48
49     return 0;
50 }
51
```